

Class Test 3

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Electricity and Electronics EBN 122

23 October 2019

MEMO

Test Information

Maximum marks:	16	Full marks:	15
Duration of test:	30 min + (10 min for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		10	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number 01 in this question paper refers to the numeric field 01 of the OCR form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

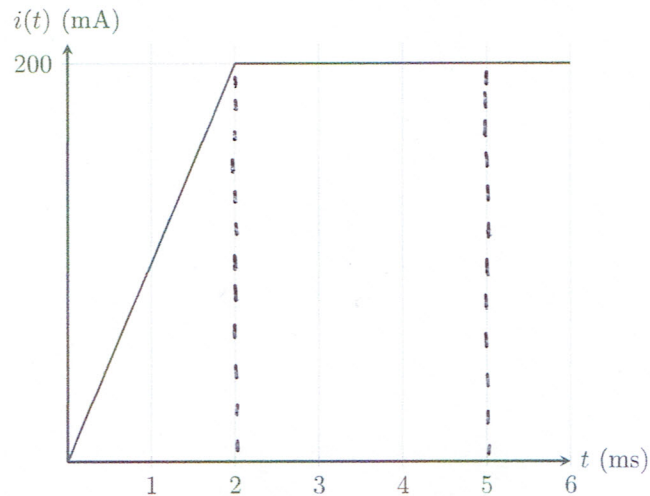
Lecturers: Prof J.P. Jacobs, Prof J.P. de Villiers, Dr X. Ye**INSTRUCTIONS****Do step 0 in the first 5 minutes of the session. (-5 to 0 min)****Do step 1 in the first 30 minutes of the test. (0-30 min)****Do step 2 within the following 5 minutes. (30-35 min)****STEP 0 (-5 min to 0 min):** Complete the front page of the OCR form.**STEP 1 (0 min to 30 min):** Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.**STEP 2 (30 min to 35 min):** (only if numeric sheets are completed) Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122C03**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

Questions 1 to 2 [2]

The current $i(t)$ is charging a 2 mF capacitor. The capacitor was not charged initially.



Calculate the voltage $v(t)$ across the capacitor at $t_1 = 2$ ms and $t_2 = 5$ ms.

01 $v(t_1)$ [1]

02 $v(t_2)$ [1]

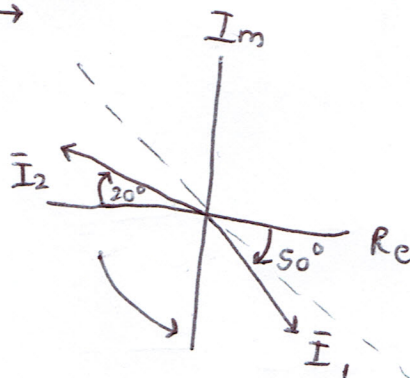
$$\begin{aligned}
 \underline{01}: v(t_1) &= \frac{1}{C} \int_0^{t_1} i(t) dt \\
 &= \frac{1}{2 \times 10^{-3}} \int_0^{2 \text{ ms}} 100t dt \\
 &= 50 \times 10^3 \left[\frac{1}{2} t^2 \right]_0^{2 \text{ ms}} = 0,1 \text{ V} = \underline{100 \text{ mV}}
 \end{aligned}$$

$$\begin{aligned}
 \underline{02}: v(t_2) &= v(t_1) + \frac{1}{C} \int_{t_1}^{t_2} i(t) dt \\
 &= 100 \text{ mV} + \int_{2 \text{ ms}}^{5 \text{ ms}} 200 \times 10^{-3} dt \times \frac{1}{2 \times 10^{-3}} \\
 &= 100 \text{ mV} + 2 \times 10^{-3} [5 \times 10^{-3} - 2 \times 10^{-3}] \cdot 100 \\
 &= \underline{400 \text{ mV}}
 \end{aligned}$$

Questions 3 to 5 [3]	
Consider the following two sinusoid waveforms:	
$i_1(t) = 8 \sin(\omega t + 40^\circ)$, and $i_2(t) = -6 \cos(\omega t - 20^\circ)$.	
03	Choose the correct statement. 1. i_1 leads i_2 . 2. i_2 leads i_1 . Write 1 if statement 1 is correct or 2 if statement 2 is correct. [1]
04	Find the phase difference $\phi_d \in [-180^\circ, 180^\circ]$ between i_1 and i_2 . [1]
05	Find $\bar{I} = \bar{I}_1 - \bar{I}_2$ in polar form. [2]

03: $i_1(t) = 8 \sin(\omega t + 40^\circ)$
 $= 8 \cos(\omega t + 40^\circ - 90^\circ)$
 $= 8 \cos(\omega t - 50^\circ)$

$\Rightarrow \bar{I}_1 = 8 \angle -50^\circ$



$\sin(x) = \cos(x - 90^\circ)$

$i_2(t) = -6 \cos(\omega t - 20^\circ)$
 $= 6 \cos(\omega t + 160^\circ)$

$\Rightarrow \bar{I}_2 = 6 \angle 160^\circ$

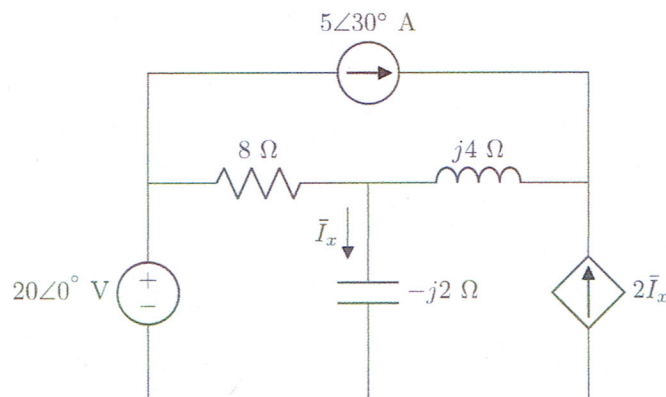
$i_1(t)$ leads $i_2(t)$

04: $\phi_d = 20^\circ + 90^\circ + 40^\circ$
 $= 150^\circ$

05: $\bar{I} = 8 \angle -50^\circ - 6 \angle 160^\circ$
 $= 13.53 \angle -37.19^\circ$

Questions 6 to 7 [4]

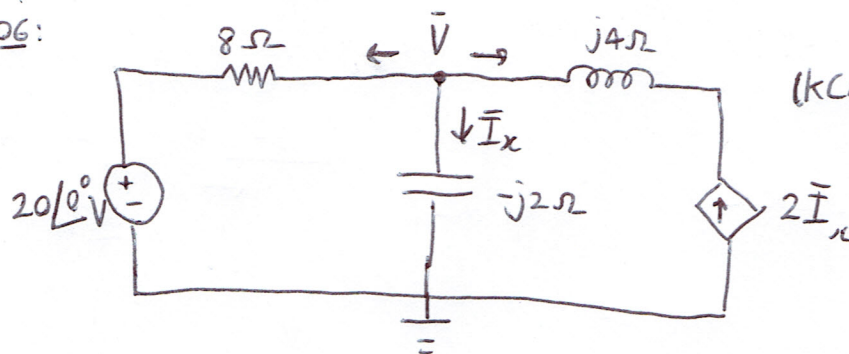
Consider the circuit diagram below.



$$\bar{I}_x = \frac{\bar{V}}{-j2}$$

06 Find the contribution to $\bar{I}_x$ from the voltage source ($\bar{I}_x^V$) in rectangular form. [2]07 Find the contribution to $\bar{I}_x$ from the independent current source ($\bar{I}_x^A$) in rectangular form. [2]

Q6:



$$(KCL): \frac{\bar{V} - 20}{8} + \frac{\bar{V}}{-j2} - 2\left(\frac{\bar{V}}{-j2}\right) = 0$$

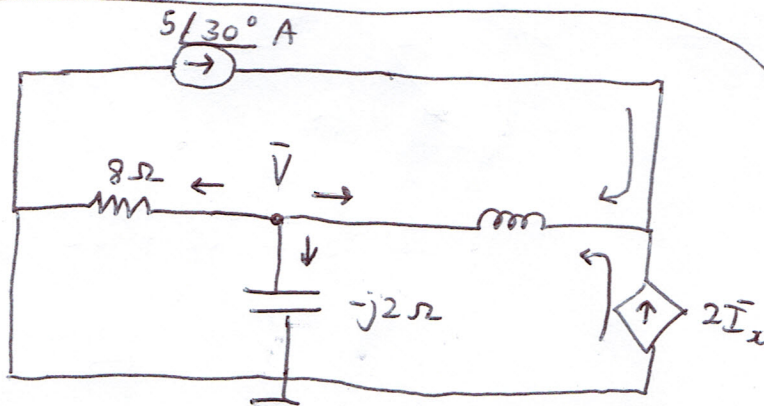
$$\bar{V} \left(\frac{1}{8} + \frac{1}{-j2} - \frac{2}{-j2} \right) = \frac{20}{8}$$

$$\bar{V} = 1,18 + j4,71 \text{ V}$$

$$\bar{I}_x^V = \frac{1,18 + j4,71}{-j2}$$

$$= -2,35 + j0,59 \text{ A}$$

Q7:



$$(KCL): \frac{\bar{V}}{8} + \frac{\bar{V}}{-j2} - \left(5\angle 30^\circ + \frac{2\bar{V}}{-j2} \right) = 0$$

$$\bar{V} \left(\frac{1}{8} + \frac{1}{-j2} - \frac{2}{-j2} \right) = 5\angle 30^\circ$$

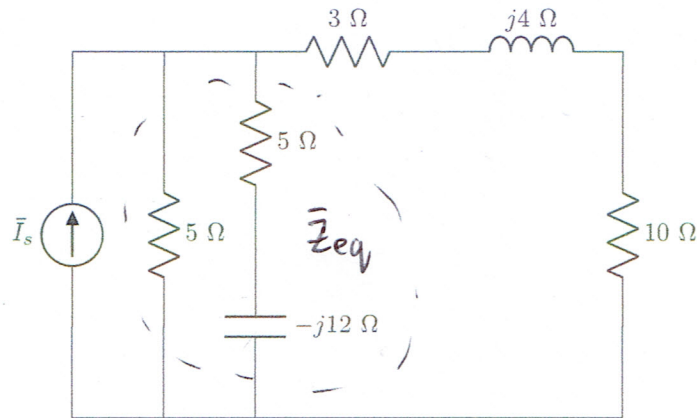
$$\bar{V} = -2,66 + j9,33 \text{ V}$$

$$\bar{I}_x^A = -4,66 - j1,33 \text{ A}$$

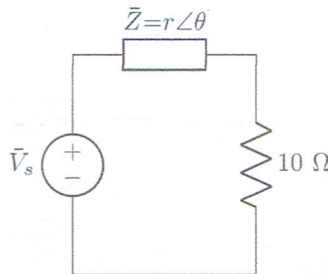
Questions 8 to 9 [3]

Consider circuits *A* and *B* below. In circuit *A*, $\bar{I}_s = j4$ A.

Circuit A:



Circuit B:



Use source transformation to convert circuit *A* to circuit *B*.

08 Find $\bar{Z}$ in polar form. [2]

09 Find $\bar{V}_s$ in polar form. [1]

Turn over for additional space for Questions 8–9.

$$\bar{Z}_{eq} = \left(\frac{1}{5} + \frac{1}{5 - j12} \right)^{-1}$$

$$= 3,975 - j1,23 \Omega$$

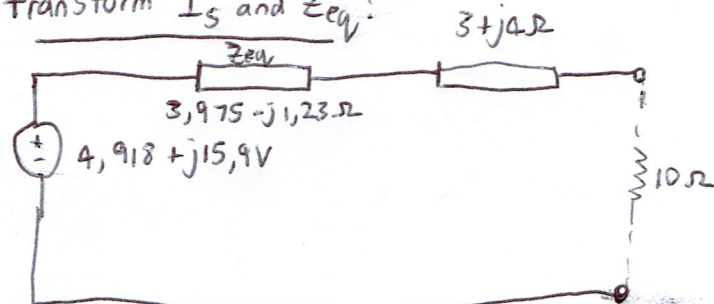
$$\bar{Z} = 3,975 - j1,23 + 3 + j4$$

$$= 6,975 + j2,77 \Omega$$

$$\rightarrow$$

$$= 7.505 \angle 21.66^\circ$$

transform $\bar{I}_s$ and $\bar{Z}_{eq}$:



$$\bar{V}_s = 4,918 + j15,9V$$

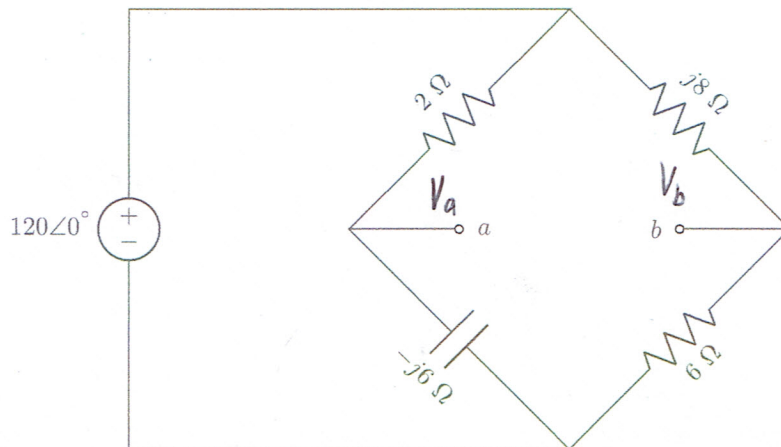
$$\rightarrow$$

$$= 16.6 \angle 72.8^\circ$$

Additional space for Questions 8–9.

Question 10 [2]

Consider the circuit below.

10 Obtain the Thévenin equivalent voltage V_{Th} at terminals $a - b$ in **rectangular** form. [2]10:

$$V_a = 120 \left(\frac{-j6}{-j6 + j2} \right)$$

$$= 108 - j36 \text{ V}$$

→

voltage division

$$V_b = 120 \left(\frac{6}{6 + j8} \right)$$

$$= 43,2 - j57,6 \text{ V}$$

→

$$V_{Th} = V_a - V_b$$

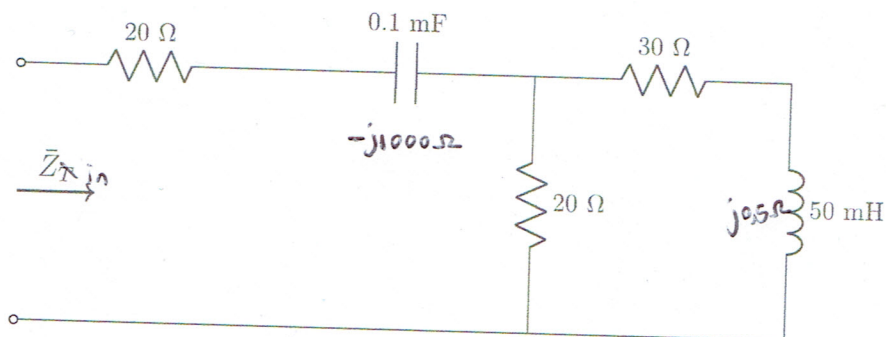
$$= 108 - j36 - (43,2 - j57,6)$$

$$= 64,8 + j21,6 \text{ V}$$

→

Question 11 [2]

Consider the circuit below.



- 11 Determine $\bar{Z}_{in}$ when $\omega = 10$ rad/s in rectangular form. [2]

$$\underline{Q11:} \quad X_C = \frac{1}{j\omega C} = \frac{1}{j(0,1 \times 10^{-3})(10)} = -j1000 \Omega$$

$$X_L = j\omega L = j0(10)(0,05) = j0,5 \Omega$$

$$\begin{aligned} \bar{Z}_{in} &= \left(\frac{1}{30 + j0,5} + \frac{1}{20 - j1000} \right)^{-1} + 20 - j1000 \\ &= 32 - j999,92 \Omega \\ &\rightarrow \end{aligned}$$

This page is intentionally left open for rough work.

Total Section 1: [16]

Grand Total : [16]

PRACTICE ANSWER SHEET

Assessment ID	2	0	1	9	E	B	N	1	2	2	C	0	3
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	Answer	Unit
01	$v(t_1) = 100$	mV
02	$v(t_2) = 400$	mV
03	Write ① or 2 in OCR form.	No units
04	$\phi_d = 150$	°
05	$I = 13,53 / -37,19^\circ$	No units
06	Contribution to I_x from voltage source = $-2,53 + j0,59$	A
07	Contribution to I_x from independent current source = $-4,66 - j1,33$	A
08	$Z = 6,975 + j2,77$	Ω
09	$V_s = 4,918 + j15,9$	V
10	$V_{Th} = 64,8 + j21,6$	V
11	$Z_{in} = 32 - j999,72$	Ω